

End-Semester Exam: Discrete Mathematics (CSE121)

Answer Key and Rubric

Date: 08-12-2025

Maximum marks: 76

Instructions. Please read the following points carefully.

- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- Write with either a black pen or a blue pen. Use of pencil is strictly prohibited.
- If you are found copying or talking to any other student, you will be disqualified from this exam, and the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or the invigilators.
- This exam is designed for 3 hours.
- Unless specified otherwise, you may freely use results proven in the lectures. However, please clearly mention whenever you invoke such a result. If you want to use any other result which is not covered in lectures, you must prove it before using it.

This is just an answer-key, so some details may be missing, which you were expected to write. Moreover, these are not the only correct solutions. If you have written a correct solution, which does not match with the answer key, you will get full marks.

Partial marks for incomplete answers and missing steps depends on the solution. You are not allowed to bargain marks for incorrect or incomplete solution. And, it shall be considered as frivolous request, penalty may be imposed accordingly.

If your answer are written in informal manner and vague manner, you will not get full marks even if your intuitions are correct. These are the just answer keys, it's not possible to make a rubric for all the solution.

Kindly review the rubric before requesting regrading. A penalty may be imposed for frivolous requests.

1 Problem Set

Question 1.1. 1. (7 marks). Let p be an arbitrary prime number and $a \in \mathbb{N}$ be an arbitrary number satisfying $a > 1$. Prove or disprove the following: $p^{\frac{1}{a}}$ is a rational number.

Solution and Rubric:

We will disprove that $p^{\frac{1}{a}}$ is a rational number.

So our claim is:

“Let p be any arbitrary prime number and a be any natural number greater than 1. Then $p^{\frac{1}{a}}$ is irrational.”

Let us assume $p^{\frac{1}{a}}$ is a rational number. Then

$$p^{\frac{1}{a}} = \frac{t}{m}$$

such that t, m are co-prime without loss of generality.

$$\Rightarrow mp^{\frac{1}{a}} = t$$

$$\Rightarrow \left(mp^{\frac{1}{a}}\right)^a = t^a$$

$$\Rightarrow m^a \cdot p = t^a \quad (1)$$

From the above expression it is clear that

$$p \mid (m^a \cdot p)$$

so

$$p \mid t^a \quad (\text{i.e. } t^a = m^a \cdot p).$$

We now prove a claim using Euclid's lemma.

Claim 1. Let t be any integer and n be any natural number. Then

$$S(n) : \text{If } p \mid t^n, \text{ then } p \mid t.$$

Proof (by induction on n):

Base case ($n = 1$). This is trivially true: if $p \mid t$ then $p \mid t$.

Induction hypothesis. Assume $S(n)$ is true for $n = k - 1$, i.e.

$$p \mid t^{k-1} \implies p \mid t.$$

Induction step. Check for $n = k$. Suppose

$$p \mid t^k.$$

Since

$$t^k = t^{k-1} \cdot t,$$

we have

$$p \mid (t^{k-1} \cdot t).$$

Using Euclid's lemma, we conclude:

$$p \mid t^{k-1} \quad \text{or} \quad p \mid t.$$

If $p \mid t$, we are done. If $p \mid t^{k-1}$, then by the induction hypothesis $p \mid t$. Thus $S(k)$ is true for all $k \in \mathbb{N}$.

$$\therefore p \mid t^n \Rightarrow p \mid t.$$

From expression (1) we obtained $p \mid t^a$, so using Claim 1 we get $p \mid t$.

Now we can write $t = up$ for some $u \in \mathbb{Z}$.

Substituting into expression (1):

$$m^a \cdot p = t^a \quad (\text{Expression 1})$$

$$\Rightarrow m^a \cdot p = (up)^a = u^a p^a$$

$$\Rightarrow m^a = u^a p^{a-1} \quad (2)$$

Given $a > 1$, we have $a - 1 > 0$, therefore $a - 1 \geq 1$.

Hence

$$p \mid u^a p^{a-1}.$$

Using expression (2),

$$p \mid m^a.$$

Now using Claim 1 again, we conclude $p \mid m$.

Thus we have proved $p \mid t$ and $p \mid m$, which contradicts our assumption that t, m are co-prime. Therefore our assumption is wrong. Hence

$p^{\frac{1}{a}}$ is irrational for any prime p and natural number $a > 1$.

Rubric:

- 1 mark for correctly identifying that we need to prove $p^{\frac{1}{a}}$ is irrational.
- 1 mark for assuming t, m are co-prime.
- 3 marks for proving the claim “if $p \mid t^a$ then $p \mid t$.”
- 2 marks for correctly arguing that $p \mid t, p \mid m$ (which contradicts t, m are co-prime).
- If a student tries to disprove using a counterexample, award at most 1 mark.

2. (6 marks). Let A, B, C be arbitrary sets contained in the universal set U . Prove the following. Here, you can not assume any result from lectures and you will have to prove them from scratch.

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(b) $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Solution and Rubric:

Let A, B, C be arbitrary sets contained in the universal set U .

(a) Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof: We prove equality by showing mutual inclusion ($\subseteq$ and $\supseteq$).

Part A: Forward Inclusion ($\subseteq$)

Let x be an arbitrary element such that:

$$x \in A \cap (B \cup C)$$

$$\implies x \in A \text{ and } x \in (B \cup C)$$

Since $x \in (B \cup C)$, there are two cases:

- **Case 1:** $x \in B$. Since $x \in A$, we have $x \in A$ and $x \in B$. Therefore $x \in (A \cap B)$.
- **Case 2:** $x \in C$. Since $x \in A$, we have $x \in A$ and $x \in C$. Therefore $x \in (A \cap C)$.

Thus in either case:

$$x \in (A \cap B) \cup (A \cap C)$$

Hence $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Part B: Backward Inclusion ($\supseteq$)

Let $x \in (A \cap B) \cup (A \cap C)$. This means at least one of the following is true:

- **Case 1:** $x \in (A \cap B)$, meaning $x \in A$ and $x \in B$. Since $x \in B$, we also have $x \in (B \cup C)$. Therefore $x \in A \cap (B \cup C)$.
- **Case 2:** $x \in (A \cap C)$, meaning $x \in A$ and $x \in C$. Since $x \in C$, we also have $x \in (B \cup C)$. Therefore $x \in A \cap (B \cup C)$.

Thus $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

Conclusion: $(A \cap B) \cup (A \cap C) = A \cap (B \cup C)$

Rubric:

- 1.5 marks for correct forward inclusion.
- 1.5 marks for correct backward inclusion.

(b) **Prove** $\overline{A \cup B} = \overline{A} \cap \overline{B}$

Proof: Part A: Forward Inclusion ($\subseteq$)

Let $x \in \overline{A \cup B}$.

- This means $x \notin (A \cup B)$.
- If x were in A or B , it would belong to the union.
- Thus $x \notin A$ and $x \notin B$.

Therefore $x \in \overline{A}$ and $x \in \overline{B}$, so:

$$x \in \overline{A} \cap \overline{B}.$$

Hence $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$.

Part B: Backward Inclusion ($\supseteq$)

Assume $x \in \overline{A} \cap \overline{B}$.

- Then $x \notin A$ and $x \notin B$.
- Therefore $x \notin (A \cup B)$.

Thus $x \in \overline{A \cup B}$. Hence $\overline{A \cup B} \supseteq \overline{A} \cap \overline{B}$

Conclusion: Mutual inclusion proves $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Rubric:

- 1.5 marks for correct forward inclusion.
- 1.5 marks for correct backward inclusion.

3. (3 marks). Let $m, n \in \mathbb{N}$. Prove or disprove: $2 \mid m^2 - n^2$.

Solution and Rubric:

We will **disprove** the statement by providing a counterexample. Take: $m = 2, n = 1$.

Check the claim

$$m^2 - n^2 = 2^2 - 1^2 = 4 - 1 = 3.$$

The number 3 is **odd**, so

$$2 \nmid (m^2 - n^2).$$

Thus, for the pair $(m, n) = (2, 1)$, the given divisibility condition does not hold.

Conclusion: Since the statement was claimed to be true for **all** natural numbers m, n , and we have found a single counterexample where

$$2 \nmid (m^2 - n^2),$$

The original statement is **false**. ■

Rubric:

- 1 mark for Provides a valid counterexample (m, n) from $\mathbb{N}$.
- 2 marks for Correctly calculates $m^2 - n^2$ and checks parity/divisibility.

Question 1.2. 1. (4 marks). Prove the following for every $n \in \mathbb{N}, n \geq 1$.

$$1 + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n + 1)! - 1,$$

where for any $a \in \mathbb{N} - \{0\}$, $a! := a \cdot (a - 1) \cdot (a - 2) \cdots 2 \cdot 1$.

Solution and Rubric:

Problem: Prove for every $n \in \mathbb{N}, n \geq 1$:

$$1 + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n + 1)! - 1.$$

Proof (by Mathematical Induction):

Step 1: Base Case ($n = 1$)

$$LHS = 1 \cdot 1! = 1$$

$$RHS = (1 + 1)! - 1 = 2! - 1 = 1$$

So the statement holds for $n = 1$.

Step 2: Inductive Hypothesis

Assume the statement is true for $n - 1$:

$$1 + 2 \cdot 2! + \cdots + (n - 1)(n - 1)! = n! - 1.$$

Step 3: Inductive Step

Consider the sum up to n :

$$S_n = (1 + 2 \cdot 2! + \cdots + (n - 1)(n - 1)!) + n \cdot n!.$$

Substitute the inductive hypothesis:

$$S_n = (n! - 1) + n \cdot n! = n! + n \cdot n! - 1.$$

Factor:

$$S_n = n!(1 + n) - 1 = (n + 1)! - 1.$$

Thus the formula holds for n .

Conclusion: By induction, the identity is true for all $n \geq 1$.

Rubric:

- 1 mark for correct base case.
- 1 mark for the correct induction hypothesis.
- 2 marks for induction step.

2. (6 marks). Let $a, b \in \mathbb{N} - \{0\}$ and p be a prime number. Suppose $p^k \mid ab$ for some $k \in \mathbb{N} - \{0\}$. Then, prove or disprove that there exists a unique $i \in \{0, 1, \dots, k\}$ such that $p^i \mid a$ and $p^{k-i} \mid b$.

Solution and Rubric:

We will **disprove** the statement by providing a counterexample that shows the value of i is not necessarily unique.

Let

$$p = 2, \quad a = 4 = 2^2, \quad b = 4 = 2^2, \quad k = 3.$$

Step 1: Check that $p^k \mid ab$

$$ab = 4 \cdot 4 = 16 = 2^4.$$

Since $2^3 = 8 \mid 16$, we have $p^k \mid ab$.

Step 2: Check the condition for different values of i

We test $i \in \{0, 1, 2, 3\}$ such that

$$2^i \mid a \quad \text{and} \quad 2^{3-i} \mid b.$$

- For $i = 1$:

$$2^1 = 2 \mid 4 = a, \quad 2^{3-1} = 4 \mid 4 = b.$$

So $i = 1$ works.

- For $i = 2$:

$$2^2 = 4 \mid 4 = a, \quad 2^{3-2} = 2 \mid 4 = b.$$

So $i = 2$ also works.

Thus, there are **two** valid values of i (namely $i = 1$ and $i = 2$), so i is **not unique**.

Conclusion: Therefore, the claim that there exists a **unique** i such that

$$p^i \mid a \quad \text{and} \quad p^{k-i} \mid b$$

is **false**. The statement is disproved by counterexample. ■

Rubric:

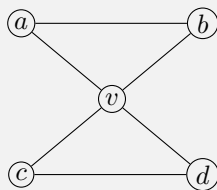
- 1 mark for stating intention to disprove via counterexample.
- 2 marks for Choosing valid example with p prime and suitable a, b, k Correctly verifies $p^k \mid ab$.
- 3 marks for finding one valid i and verifies both divisibility conditions Finds a second valid i and verifies both divisibility conditions

Question 1.3. 1. (3.5 marks). Define Euler tour and Hamiltonian cycle in an undirected graph. Prove or disprove: every Euler tour is also a Hamiltonian cycle.

Solution and Rubric:

Euler tour / Eulerian circuit: An Euler tour (or Eulerian circuit) of an undirected graph G is a closed trail that uses every edge of G exactly once. A connected undirected graph admits an Euler tour iff every vertex has even degree.

Hamiltonian cycle: A Hamiltonian cycle in an undirected graph G is a cycle which visits every vertex of G exactly once and returns to the start vertex.



Claim: Every Euler tour is not necessarily a Hamiltonian cycle.

Proof. We disprove the statement by counterexample: we give an undirected graph that has an Euler tour but has no Hamiltonian cycle.

Consider the graph G consisting of two triangles that share exactly one vertex. Concretely, let the vertex set be

$$V(G) = \{v, a, b, c, d\},$$

and the edge set be

$$E(G) = \{va, ab, bv, vc, cd, dv\}.$$

So v, a, b form a triangle and v, c, d form another triangle; the two triangles meet only at v .

(1) G has an Euler tour. The degrees are

$$\deg(v) = 4, \quad \deg(a) = \deg(b) = \deg(c) = \deg(d) = 2,$$

so every vertex has even degree and G is connected. By the classical criterion for Eulerian graphs, G admits an Euler tour (for instance one Eulerian circuit is

$$v-a-b-v-c-d-v,$$

which uses each edge exactly once).

(2) G has no Hamiltonian cycle. Suppose, for contradiction, that C is a Hamiltonian cycle in G . A Hamiltonian cycle must visit each of the five vertices v, a, b, c, d exactly once and be a cycle. Observe that removing vertex v disconnects G into two components $\{a, b\}$ and $\{c, d\}$. In any cycle that contains all vertices exactly once, the removal of the vertex v would leave a single path joining the two neighbors of v along the cycle. But since v is the only vertex joining the $\{a, b\}$ part to the $\{c, d\}$ part, any cycle that contains vertices from both parts must

pass through v at least twice to switch from the $\{a, b\}$ -side to the $\{c, d\}$ -side. This contradicts the requirement that a Hamiltonian cycle visits v exactly once. Therefore no Hamiltonian cycle exists in G .

Because G admits an Euler tour but no Hamiltonian cycle, the claim “every Euler tour is also a Hamiltonian cycle” is false. $\square$

Remark. Intuitively, an Euler tour cares about edges (it must use every edge exactly once) while a Hamiltonian cycle cares about vertices (it must visit every vertex exactly once). These are orthogonal requirements: an Eulerian graph can force repeated visits of some vertices (as in the example above), while a Hamiltonian cycle is not required to use all edges. Thus neither property implies the other in general.

Rubric:

- 0.75 marks for each correct definition.
- 1 mark for diagram of counter example.
- 1 mark for the justification of counter example.

2. (3.5 marks). How many words can be obtained by permuting the characters of the following word - OPERATIONSINDOOR? Justify your answer.

Solution and Rubric:

A permutation of the letters corresponds to a linear arrangement of the 16 letter-positions. If all 16 letters were distinct there would be $16!$ arrangements. However, when some letters are identical, reordering the identical letters among themselves does not produce a new distinct word. For each letter that appears m_j times, there are $m_j!$ internal reorderings that leave the word unchanged. Hence we divide $16!$ by the product of these $m_j!$ terms, yielding the multinomial expression

$$\frac{16!}{\prod_j m_j!},$$

which reduces to the formulas given above for the two case-interpretations.

Here, the multiset of letters (length still 16) has multiplicities

$$O : 4, \quad R : 2, \quad I : 2, \quad N : 2, \quad P : 1, \quad E : 1, \quad A : 1, \quad T : 1, \quad S : 1, \quad D : 1.$$

By the Bookkeeper rule, the number of distinct permutations is

$$\frac{16!}{4! 2! 2! 2!}.$$

Rubric:

- 1.5 mark for correct answer.
- 1 mark for mentioning bookkeeper rule or subset split rule.
- 1 marks for correct justification.

3. (3 marks). Let $G = (V, E)$ be an undirected finite graph such that $|V|$ is odd (G need not be a complete graph). Prove or disprove: G has a perfect matching.

Solution and Rubric:

Disproving: G has a perfect matching. Suppose, for contradiction, that $G = (V, E)$ has a perfect matching M . By definition of a perfect matching, the edges of M pair up the vertices of V so that each vertex is an endpoint of exactly one edge in M . Each edge covers exactly two distinct vertices, hence the total number of vertices covered by M is $2|M|$. Because M is

perfect it covers all vertices of G , so

$$|V| = 2|M|.$$

The right-hand side $2|M|$ is even, so $|V|$ must be even. This contradicts the hypothesis that $|V|$ is odd. Therefore G cannot have a perfect matching.

Rubric:

- 1 mark for correct answer statement of disproving.
- 2 mark for correct justification.

Question 1.4. 1. (6 marks). Let $F : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function satisfying $F(x+y) = F(x) + F(y)$ for every $x, y \in \mathbb{Z}$. Prove that F is injective if and only if $F(1) \neq 0$.

Solution and Rubric:

Properties of the Function F : Given condition $F(x+y) = F(x) + F(y)$.

Property 1: $F(0)$

Set $x = 0$ and $y = 0$:

$$F(0+0) = F(0) + F(0).$$

This gives $F(0) = 2F(0)$, so subtracting $F(0)$ from both sides yields $F(0) = 0$.

Property 2: $F(-x)$

Set $y = -x$:

$$F(x + (-x)) = F(x) + F(-x).$$

Since $F(0) = 0$, we get

$$0 = F(x) + F(-x),$$

hence

$$F(-x) = -F(x).$$

Property 3: Explicit Formula for $F(n)$:

Claim:

$$F(n) = c \cdot n, \quad \text{where } c = F(1).$$

For each positive integer n :

Base case: $F(1)$ is some integer constant.

Inductive step: Assume $F(k) = k \cdot F(1)$. Then

$$F(k+1) = F(k) + F(1) = k \cdot F(1) + F(1) = (k+1) \cdot F(1).$$

Thus $F(n) = n \cdot F(1)$ for all positive integers n .

For negative integers n , write $n = -m$ with $m > 0$. Then

$$F(n) = F(-m) = -F(m) = -(m \cdot F(1)) = n \cdot F(1).$$

Hence, for all $n \in \mathbb{Z}$:

$$F(n) = c \cdot n, \quad \text{where } c = F(1).$$

$\Rightarrow$ **If F is injective, then $F(1) \neq 0$**

We prove the contrapositive: if $F(1) = 0$, then F is not injective.

If $F(1) = 0$, then from $F(n) = n \cdot F(1)$ we obtain:

$$F(n) = 0 \quad \text{for every integer } n.$$

Thus $F(1) = 0$ and $F(2) = 0$, even though $1 \neq 2$. Therefore F is not injective.

Hence, if F is injective, we must have $F(1) \neq 0$.

$\Leftarrow$ **If $F(1) \neq 0$, then F is injective**

Assume $F(1) = c$ where $c \neq 0$. Then $F(n) = c \cdot n$ for all integers n .

Suppose $F(n_1) = F(n_2)$. Then:

$$cn_1 = cn_2,$$

$$c(n_1 - n_2) = 0.$$

Since $c \neq 0$, dividing gives $n_1 - n_2 = 0$, so $n_1 = n_2$.

Therefore F is injective.

Conclusion: We have shown both directions:

- If F is injective, then $F(1) \neq 0$.
- If $F(1) \neq 0$, then F is injective.

Thus F is injective if and only if $F(1) \neq 0$.

Rubric (6 Marks)

- $\Rightarrow$: If F is injective $\Rightarrow F(1) \neq 0$ 3 marks
- $\Leftarrow$: If $F(1) \neq 0 \Rightarrow F$ is injective 3 marks

2. (4 marks). Let $n, d \in \mathbb{N} - \{0\}$ and $x_1, \dots, x_n$ be variables. A monomial of degree d in these n variables looks like $x_1^{e_1} x_2^{e_2} \dots x_n^{e_n}$, where $e_1, \dots, e_n \in \mathbb{N}$, $e_1 + \dots + e_n = d$. For example, $x_1^2 x_2^4 x_3$ has degree 7. Variables in every monomial have commutativity with respect to multiplication.

How many monomials of degree exactly d are there in n variables? Justify your answer.

Solution and Rubric:

We solve this problem using the bijection rule. Let A be the set of sequences of d degree monomials with n variables. To apply the bijection rule, we first have to come up with a set whose cardinality is easy to find and then give a bijection function from A to this set.

Let B be the set of $n + d - 1$ -bit 0,1-sequences containing exactly $n - 1$ 1s. We have to first find $|B|$ and then give a bijection function from A to B .

Clearly by subset rule,

$$|B| = \binom{d + n - 1}{n - 1}$$

Now we give a bijection from A to B :

An example of an element of A is $x_1^2 x_2 x_3^3 \dots x_n^2$, where the sum of powers of all x_i 's is equal to d . It can be represented as follows.

$$\underbrace{x_1}_2 \quad \underbrace{x_2}_1 \quad \underbrace{x_3}_3 \quad \dots \quad \underbrace{x_n}_2$$

We can follow represent this as follows:

$$\underbrace{00}_{x_1} 1 \underbrace{0}_{x_2} 1 \underbrace{000}_{x_3} 1 \dots 1 \underbrace{00}_{x_n}$$

We have used the space to separate two consecutive variables with 1s. We can also use some symbol other than 0 in the place of space and we specifically use 1. Then, the above element of A looks as

$$001010001 \dots 100$$

Note that it is an element of B . Note that if we map every element of A to B like this, we get all the elements of B and this map is bijective function. Then, by bijection rule and the subset rule,

$$|A| = \binom{d + n - 1}{n - 1}$$

Note: You can also use same kind of argument taught in the class for sweets example.

Rubric:

- 1 mark for correct answer.
- 3 marks for correct justification (mentioning of the bijection rule and the subset rule is crucial).

Question 1.5. (10 marks). An infinite 0,1 sequence $\mathbf{a} := a_1a_2a_3\cdots$ is said to be ‘special’ when 1’s in $\mathbf{a}$ can only occur at positions which are either prime numbers or perfect cubes of natural numbers¹. Formally, $\mathbf{a}$ is a special 0,1 sequence whenever for every $i \in \mathbb{N} - \{0\}$,

$$a_i = \begin{cases} 0 & \text{if } i \text{ is not a perfect cube and } i \text{ is not a prime} \\ 0 \text{ or } 1 & \text{if } i \text{ is a perfect cube or } i \text{ is a prime} \end{cases}$$

E.g., 00000 $\cdots$, 01100 $\cdots$, etc. are special 0,1 sequences, whereas 00010 $\cdots$ is not a special 0,1 sequence as it contains 1 at position 4, which is neither a prime nor a perfect cube. In these three examples, ‘ $\cdots$ ’ means that all the bits after the fifth bit are 0.

Let $\mathcal{A}$ be the set of all special 0,1 infinite sequences. Prove that $\mathcal{A}$ is uncountable.

Solution and Rubric:

Given

$$\mathcal{A} = \{\text{all special 0,1 infinite sequences}\}.$$

1) Let us define a function

$$f : \mathcal{A} \rightarrow B,$$

where B is the set of all infinite subsequences of $\mathcal{A}$ obtained by taking only the entries of a special sequence whose indices are primes or perfect cubes. For example, if $a_1a_2a_3a_4a_5a_6a_7\cdots$ is a sequence in $\mathcal{A}$, then its counterpart in B is $a_1a_2a_3a_5a_7\cdots$. Note that we have removed the elements indexed by 4 and 6 and so on, as these indices are neither primes nor perfect cubes.

Let $\mathbf{a} \in \mathcal{A}$. Write

$$\mathbf{a} = a_1a_2a_3a_4a_5a_6\cdots$$

Define

$$f(\mathbf{a}) = a_1a_2a_3a_5a_7a_8\cdots,$$

that is, the sequence consisting only of entries of $\mathbf{a}$ whose indices are prime or perfect cube numbers. We show that f is a bijection.

Claim 1: f is one-one

Suppose

$$f(a_1a_2a_3a_4a_5a_6\cdots) = f(b_1b_2b_3b_4b_5b_6\cdots).$$

This means all prime and perfect-cube-indexed entries of the first sequence equal the corresponding entries of the second sequence.

We know that every special sequence has 0 at all non-prime and non-perfect-cube positions. Thus, we immediately get that

$$a_1a_2a_3a_4a_5a_6\cdots = b_1b_2b_3b_4b_5b_6\cdots,$$

so f is injective.

Claim 2: f is onto

Given any sequence in B , we can insert zeros at all non-prime and non-cube positions and obtain a preimage in $\mathcal{A}$. Hence f is onto.

Therefore, f is a bijection from $\mathcal{A}$ to B .

2) Let

$$P := \text{set of all prime numbers}, \quad C := \text{set of all perfect cube}.$$

Clearly, $P \cup C \subseteq \mathbb{N}$ and we know that P and C are countably infinite. Hence, $P \cup C$ is also countably infinite. Thus, there exists a bijection $\mathbb{N} \rightarrow P \cup C$. Write

$$P \cup C = \{v_1, v_2, v_3, v_4, \dots\} = \{1, 2, 3, 5, 7, 8, \dots\},$$

where $v_1v_2, \dots$ is an enumeration of $P \cup C$ that we are taking into consideration. Define a function

$$g : B \rightarrow \text{pow}(P \cup C) \quad (\text{the power set of } P \cup C)$$

¹A natural number x is called a perfect cube if and only if there exists $y \in \mathbb{N}$ such that $x = y^3$. For example, 0,1,8,27 are perfect cubes.

as follows. For $\mathbf{b} = b_{v_1}b_{v_2}b_{v_3}\cdots \in B$, define

$$g(\mathbf{b}) = \{v_i : b_{v_i} = 1, v_i \in P \cup C\}.$$

Claim 3: g is a bijection.

Proof (one-one). Suppose $g(\mathbf{b}) = g(\mathbf{b}')$ for some $\mathbf{b} = b_{v_1}b_{v_2}b_{v_3}\cdots, \mathbf{b}' = b'_{v_1}b'_{v_2}b'_{v_3}\cdots \in B$. Then, for every $v_i \in P \cup C, b_{v_i} = b'_{v_i}$. Let

$$S_{\mathbf{b}} = \{v_i : b_{v_i} = 1, v_i \in P \cup C\}, S_{\mathbf{b}'} = \{v_i : b'_{v_i} = 1, v_i \in P \cup C\}.$$

Clearly, $S_{\mathbf{b}}, S_{\mathbf{b}'} \subseteq P \cup C$ and from above we get

$$S_{\mathbf{b}} = S_{\mathbf{b}'}.$$

This implies g is injective.

Proof (onto). Let $S \subseteq (P \cup C)$. Define a sequence $\mathbf{b} = b_{v_1}b_{v_2}b_{v_3}\cdots$ by

$$b_i = \begin{cases} 1, & v_i \in S, \\ 0, & v_i \notin S. \end{cases}$$

Clearly, $\mathbf{b} \in B$ and $g(\mathbf{b}) = S$. Hence every subset of $P \cup C$ has a pre-image, so g is onto.

Therefore g is a bijection.

Thus we have bijection

$$f : A \rightarrow B, \quad g : B \rightarrow \text{pow}(P \cup C).$$

3) To show that A is uncountable, it is enough to show that B is uncountable, which will follow from showing uncountability of $\text{pow}(P \cup C)$. As P is infinite, it follows that the power set $\text{pow}(P \cup C)$ is also infinite. For contradiction, let us now assume that $\text{pow}(P \cup C)$ is countably infinite. Then, by definition, there exists a bijection $\mu : \mathbb{N} \rightarrow \text{pow}(P \cup C)$. As noted above, $P \cup C$ is countably infinite, and thus, we know that there exists a bijective function from $\mathbb{N}$ to $P \cup C$. Thus, as per our assumption that $\text{pow}(P \cup C)$ is countably infinite, there exists a bijective function $h : P \cup C \rightarrow \text{pow}(P \cup C)$. As $\text{pow}(P \cup C)$ is countably infinite, we can enumerate its elements as

$$\text{pow}(P \cup C) = \{S_1, S_2, S_3, \dots\}.$$

Since $P \cup C = \{v_1, v_2, v_3, \dots\}$, each subset S_i has a characteristic function

$$\chi_{S_i} : (P \cup C) \rightarrow \{0, 1\}$$

defined by

$$\chi_{S_i}(v_m) = \begin{cases} 0, & v_m \notin S_i, \\ 1, & v_m \in S_i. \end{cases}$$

We form the table of characteristic values:

	v_1	v_2	v_3	v_4	$\cdots$
χ_{S_1}	a_{11}	a_{12}	a_{13}	a_{14}	$\cdots$
χ_{S_2}	a_{21}	a_{22}	a_{23}	a_{24}	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

where

$$a_{ij} = \begin{cases} 0, & v_j \notin S_i, \\ 1, & v_j \in S_i. \end{cases}$$

Now define a new subset $D \subseteq (P \cup C)$ by specifying its characteristic function along the diagonal, i.e., for every $v_i \in P \cup C$

$$\chi_D(v_i) = 1 - \chi_{S_i}(v_i).$$

Claim 4: $\chi_D \neq \chi_{S_i}$ for every $i \in \mathbb{N}$.

If, for contradiction, $\chi_D = \chi_{S_j}$ for some j , then looking at the coordinate v_j gives

$$\chi_D(v_j) = \chi_{S_j}(v_j).$$

But by definition of χ_D ,

$$\chi_D(v_j) = 1 - \chi_{S_j}(v_j),$$

so we obtain the impossible equality

$$1 - \chi_{S_j}(v_j) = \chi_{S_j}(v_j) \Rightarrow 1 = 0.$$

Thus D is a subset of $P \cup C$ whose characteristic function does not appear in the table above. Therefore there does not exist a $j \in P \cup C$ satisfying $h(j) = D$. This means that

$$D \notin \text{range}(h),$$

so h is not onto and hence not a bijection. This contradicts the assumption that $\text{pow}(P \cup C)$ is countably infinite. Therefore,

$$\text{pow}(P \cup C) \text{ is uncountable.}$$

Since $f : A \rightarrow B$ and $g : B \rightarrow \text{pow}(P \cup C)$ are bijections, we conclude that A is uncountable.

Rubric:

- 2 marks for constructing the bijection map $f : A \rightarrow B$.
- 1 mark for correctly showing that $P \cup C$ is countably infinite.
- 2.5 marks for constructing the bijection map $g : B \rightarrow \text{pow}(P \cup C)$.
- 4.5 marks for proving that $\text{pow}(P \cup C)$ is uncountable.

Question 1.6. 1. (8 marks). Let $n \in \mathbb{N}, n \geq 3$. Let $A = \{1, \dots, n\}$. Consider the following relation R on $\text{pow}(A)$: for every $X, Y \in \text{pow}(A)$, $(X, Y) \in R$ if and only if for every $Z \subseteq A, |Z| = 2$, we have $|X \cap Z| = |Y \cap Z|$. Is R an equivalence relation? If yes, how many equivalence classes are there? Justify your answer. What happens to the number of equivalence classes for $n = 2$?

Solution and Rubric:

Proof that R is an equivalence relation

Recall that

$$(X, Y) \in R \iff \forall Z \subseteq A, |Z| = 2, |X \cap Z| = |Y \cap Z|.$$

Reflexivity For every $X \in \text{pow}(A)$ and every $Z \subseteq A$ with $|Z| = 2$,

$$|X \cap Z| = |X \cap Z|.$$

Hence $(X, X) \in R$. Therefore R is reflexive.

Symmetry Assume $(X, Y) \in R$. Then for every $Z \subseteq A$ with $|Z| = 2$,

$$|X \cap Z| = |Y \cap Z|.$$

Hence for every $Z \subseteq A$ with $|Z| = 2$

$$|Y \cap Z| = |X \cap Z|,$$

so $(Y, X) \in R$. Therefore R is symmetric.

Transitivity Assume $(U, X) \in R$ and $(X, Y) \in R$. Then for every $Z \subseteq A$ with $|Z| = 2$,

$$|U \cap Z| = |X \cap Z| \quad \text{and} \quad |X \cap Z| = |Y \cap Z|.$$

Hence

$$|U \cap Z| = |Y \cap Z|.$$

Hence for every $Z \subseteq A$ with $|Z| = 2$ Thus $(U, Y) \in R$. Therefore R is transitive.

Since R is reflexive, symmetric, and transitive, it is an equivalence relation.

Number of equivalence classes (for $n \geq 3$)

We next determine the equivalence classes of R . A crucial first observation is the following.

The empty set forms its own equivalence class

If $X = \emptyset$ and $Y \neq \emptyset$, choose any element $a \in Y$. Because $n \geq 3$, we can pick a second distinct element $b \in A \setminus Y$. Let $Z = \{a, b\}$. Then

$$|\emptyset \cap Z| = 0, \quad |Y \cap Z| = 1.$$

Thus $(\emptyset, Y) \notin R$.

Therefore,

$\emptyset$ is not related to any non-empty set.

Hence $\{\emptyset\}$ is an equivalence class by itself.

Case Analysis for non-empty sets

From now on, assume both X and Y are non-empty subsets of A .

Case 1: $|X| = |Y|$ but $X \neq Y$

At least one element must differ between them.

Subcase 1a: $|X| = |Y| = 1$. Let $X = \{a\}$, $Y = \{b\}$ with $a \neq b$. Since $n \geq 3$, choose $c \in A \setminus \{a, b\}$. Set $Z = \{a, c\}$. Then

$$|X \cap Z| = 1, \quad |Y \cap Z| = 0.$$

Thus $(X, Y) \notin R$.

Subcase 1b: $|X| = |Y| > 1$. Since $X \neq Y$, choose $a \in X \setminus Y$. Because X is non-empty and $|X| > 1$, choose another $d \in X$ with $d \neq a$. Let $Z = \{a, d\}$. Then

$$|X \cap Z| = 2, \quad |Y \cap Z| = 1.$$

Thus $(X, Y) \notin R$.

Conclusion for Case 1. If $|X| = |Y|$ but $X \neq Y$, and both are non-empty, then $(X, Y) \notin R$. Thus no two different non-empty sets of equal size are equivalent.

Case 2: $|X| \neq |Y|$ (both sets non-empty)

WLOG Assume $|X| > |Y|$. Pick $a \in X \setminus Y$. Since X is non-empty and $|X| \geq 2$, choose another $d \in X$. Let $Z = \{a, d\}$. Then

$$|X \cap Z| = 2, \quad |Y \cap Z| \leq 1.$$

Thus $(X, Y) \notin R$.

Conclusion for $n \geq 3$.

Combining all cases and the fact that the empty set is not related to any non-empty set:

$$(X, Y) \in R \iff X = Y.$$

Thus each subset of A forms its own equivalence class. Therefore,

$$\text{number of equivalence classes} = |\text{pow}(A)| = 2^n.$$

Case: $n = 2$

Let $A = \{1, 2\}$. The only subset $Z \subseteq A$ with $|Z| = 2$ is A itself. Thus

$$(X, Y) \in R \iff |X \cap A| = |Y \cap A|.$$

Since $X \cap A = X$ and $Y \cap A = Y$, we have

$$(X, Y) \in R \iff |X| = |Y|.$$

Hence the equivalence classes are

$$\{\emptyset\}, \quad \{\{1\}, \{2\}\}, \quad \{\{1, 2\}\}.$$

Thus when $n = 2$, there are exactly 3 equivalence classes. □

Rubric (8 Marks):

- 3 marks for proving that R is an equivalence relation:
 - 1 mark for reflexivity,
 - 1 mark for symmetry,
 - 1 mark for transitivity.
- 4 marks for determining the number of equivalence classes when $n \geq 3$:
 - 1 mark for analysing the case of the empty set $\emptyset$,
 - 1 mark for analysing the case where both sets have the same cardinality,
 - 1 mark for analysing the case where the sets have different cardinalities,
 - 1 mark for concluding that the number of equivalence classes is 2^n .
- 1 mark for computing the number of equivalence classes in the special case $n = 2$.

2. (8 marks). Let $n, r \in \mathbb{N} - \{0\}, r \leq n$. Give a combinatorial proof for the following identity. You will not get any mark in this part if you give an algebraic proof. Also, you must clearly mention the combinatorial rule at every step of your proof.

$$\sum_{k=0}^n \binom{k}{r} = \binom{n+1}{r+1}.$$

(Hint: For $k < r, \binom{k}{r} = 0$.)

Solution and Rubric:

Claim: For all $n, r \in \mathbb{N}$ with $r \leq n$,

$$\sum_{k=0}^n \binom{k}{r} = \binom{n+1}{r+1}.$$

by hint one can say that

$$\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}.$$

Proof: We prove this identity using a combinatorial (double counting) argument.

Consider a set of $n+1$ distinct objects labelled as

$$\{a_1, a_2, a_3, \dots, a_{n+1}\}.$$

We want to count the number of ways to select a subset of size $r+1$ from this set.

Method 1 (Subset Rule):

The number of ways to choose $r + 1$ objects from $n + 1$ objects is given by the combination rule as

$$\binom{n+1}{r+1}.$$

Method 2 (Partition by Largest Element):

We now count the same selections in another way. We partition all subsets of size $r + 1$ according to their largest element.

Let the largest element in the selected subset be $k + 1$, where

$$r \leq k \leq n.$$

Once $k + 1$ is fixed, the remaining r elements must be chosen from the set

$$\{1, 2, \dots, k\}.$$

By the combination rule, the number of ways to do this is

$$\binom{k}{r}.$$

Since the values of k range from r to n and each case is mutually exclusive, by the addition rule, the total number of subsets of size $r + 1$ is

$$\sum_{k=r}^n \binom{k}{r}.$$

Conclusion (Double Counting Principle):

Both methods count the same collection of subsets of size $r + 1$. Therefore, by the double counting principle, we conclude that

$$\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}.$$

■

Rubric:

- 2 marks for correct interpretation of RHS.
- 4 marks for correct interpretation of LHS.
- 2 marks for mentioning all the necessary rules.

3. (4 marks). Prove that the set of all even integers is countably infinite. You can not use any result from lectures/assignments/tutorials to prove this and you will have to prove it from scratch.

Solution and Rubric:

Claim: The set of all even integers

$$\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

is countably infinite.

Proof:

To prove that $\mathbb{Z}$ is countably infinite, we construct a bijection between the set of natural numbers $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and the set $\mathbb{Z}$.

Step 1: Definition of the function

Define the function $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$f(n) = \begin{cases} 2k, & n = 2k - 1, k \in \mathbb{N}, \\ -2k, & n = 2k, k \in \mathbb{N}. \end{cases}$$

This generates the sequence

$$0, 2, -2, 4, -4, 6, -6, \dots$$

which contains all even integers.

Step 2: Injectivity Let $n_1, n_2 \in \mathbb{N}$ and suppose $f(n_1) = f(n_2)$. We show $n_1 = n_2$.

Observe that the image of an odd argument is a positive even integer, while the image of an even argument is a negative even integer. Thus $f(n)$ is never zero and its sign determines the parity of n .

Case 1: $f(n_1) > 0$. Then n_1 is odd, so there exists $k_1 \in \mathbb{N}$ with $n_1 = 2k_1 - 1$ and $f(n_1) = 2k_1$. Since $f(n_1) = f(n_2) > 0$, n_2 must also be odd; write $n_2 = 2k_2 - 1$ with $k_2 \in \mathbb{N}$. Then

$$2k_1 = f(n_1) = f(n_2) = 2k_2,$$

hence $k_1 = k_2$ and therefore $n_1 = 2k_1 - 1 = 2k_2 - 1 = n_2$.

Case 2: $f(n_1) < 0$. Then n_1 is even, so there exists $k_1 \in \mathbb{N}$ with $n_1 = 2k_1$ and $f(n_1) = -2k_1$. As $f(n_2) = f(n_1) < 0$, n_2 is also even; write $n_2 = 2k_2$ with $k_2 \in \mathbb{N}$. Then

$$-2k_1 = f(n_1) = f(n_2) = -2k_2,$$

so $k_1 = k_2$ and hence $n_1 = 2k_1 = 2k_2 = n_2$.

Since in both possible sign-cases $n_1 = n_2$, we conclude that f is injective.

Step 3: Surjectivity

Let $e \in \mathbb{Z}$ be an arbitrary even integer.

- If $e = 0$, then $f(0) = 0$.
- If $e = 2k > 0$, then $f(2k - 1) = 2k = e$.
- If $e = -2k < 0$, then $f(2k) = -2k = e$.

Thus, for every $e \in \mathbb{Z}$, there exists $n \in \mathbb{N}$ such that $f(n) = e$. Hence, f is surjective.

Conclusion:

Since f is both injective and surjective, it is a bijection from the natural number to all even integers. Therefore, the set of all even integers is **countably infinite**. ■

Rubric (4 Marks):

- 1 mark for the correct definition of function.
- 1.5 mark for injectivity.
- 1.5 mark for surjectivity.